Bibhu Bhushan Sinha

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Senior Data Engineer | Big Data Engineer | Spark & Cloud Specialist

Notice Period- **1 month**

Core Competencies

Agile/ Scrum/Waterfall Methodologies

Data Analysis & Development

Big Data Spark Development and Data Engineering

Code Deployment & Monitoring

Spark Streaming Application using Spark-Kafka(IOT)

DOMAIN:
Cloud Services, Banking(BFSI),Aerospace & GIS, Insurance,IOT

Profile Summary

- 13+ years of experience in developing Big Data Spark application using Spark-Scala ,Pyspark.
- Oriented professional with 9+ years of experience in data engineering field as part of developing, supporting & testing the cloud based BigData/Spark products & Application
- 5 years of current working experience with agile methodology in banking & Insurance ,IOT domain.
- 1 year in Core Java and 5+ years of working experience on Cloudera Hadoop Distribution.
- Proficient in architecture of Hadoop Spark echo system like Sqoop, HDFS, YARN, Hive, Spark, SparkSQL, Scala, Pyspark along with Kafka
- Load and transform large sets of structured, semi structured data and Implemented Partitioning, Dynamic Partitions, Buckets in HIVE.
- Implemented the ingestion and processing streaming data using Kafka-Spark,Cassandra,Elastic Search(Kibana), ADLS, MQTT, Streamsets, Python REST API's , AWS(Glue,Athena,S3,DynamoDb,Step Functions,Lamda,Terraform etc), Databricks.
- Experience in Daily production support to monitor and trouble shoots Spark/Hive jobs
- Experienced in preparing and executing test cases that required for system integration testing based on user acceptance criteria.
- Depth experience in invoking jobs using Airflow Scheduler, Autosys, Control M, Tivoli.
- Responsible for deployments, performance tuning, monitoring and diagnosing servers within the big-data cloud based platforms.

IT Skills

- Technologies Used :- Spark/Pyspark, Airflow, Control-M, Autosys,Tivoli, Hadoop, Hive, Sqoop, Kafka, Azure , AWS , Databricks, Streamsets, MQTT, REST API's.
- Languages Used :-Scala, Python, Java.
- Data Visualization:- Kibana.
- SQL/No SQL DB :- postgreSQL, HBase, Mongo dB, Snowflake,Cassandra, Elastic Search, MYSQL.
- Development/UI Tools :- Eclipse, IntelliJ, SSMS,Databricks.
- Testing Methodology : -Manual/Functional testing, Integration testing, API testing.
- CI/CD & Version Control Tool :- Jenkins, Git ,SVN, Team City, Azure Devops.
- Incident management & monitoring tool :- Service Now
- Big Data Ecosystem :- ADLS, HDFS, Hive, Sqoop, Flume, Oozie, Spark, Databricks, AWS services(Glue,Athena,S3,Lamda,Step Functions,Terraform, Cloudwatch, SNS, IAM etc)

Organizational Experience

Organization: **JP Morgan Chase**
Duration: **Sep'24 - Present**
Sep'24 to Till Now with JP Morgan Chase as Software Engineer III -Data Engineering

Key Result Areas:
OSKAR is a tool for providing business users with necessary Client Onboarding, reporting, metrics and analytics via business reports and data visualization screens.

- Modernized traditional ETL workflows by reengineering them on the Databricks platform.
- Designed and developed data pipelines to ingest data from diverse sources loading into Delta tables in AWS Cloud

using Databricks and Apache NiFi.

- Implemented the Medallion Architecture (Bronze, Silver, Gold layers) for all data sources, ensuring robust data transformation processes up to the gold layer for analytical consumption.
- Built hybrid data pipelines to handle data for restricted countries, applying masking for PI attributes.
- Developed automated data reconciliation scripts in Python to compare record counts and validate primary key consistency between source systems and the bronze layer.
- Actively participated in Agile development within a Scrum framework, contributing to backlog grooming, user story creation, and sprint planning sessions.

Organization: Capgemini UK Plc
Duration: Sep'22 – Aug'24
Sep'22 to Aug'24 with Capgemini UK Plc as Consultant Data Engineering

Key Result Areas: With Client - Barclays (Sept 2022 – Aug 2024)

- Worked on creating Datamart for Investement Banking Project.Used AWS cloudformation for deploying automated infrastructure services for the data pipleines.
- Implemented robust data pipeline using AWS Glue-Pyspark and orchestration using AWS Step Functions involving Dynamo DB for maintaining metadata, job status with different file format csv,parquet etc.
- Processed,cleaned data is then ingested into AWS Redshift dataware house for multiple report generation and exposing the data to different consumer either through views or by creating dashboards depending on the requirement.
- Implemented Lambda function for updating task status , unzipping files based on S3 event notification and Event Bridge Rules for triggering Step Function on timely basis.s
- Involved in the development of data pipeline using Pyspark and orchestration using Autosys/Tivoli which picks data from different sources(EDP) and lands into SQL-HRDW regions for batch & API consumption.
- Developed staging and parsing framework which consist of reading data from GSAP/EDP regions connectivity through Pyspark engine.

Organization: LNT Infotech
Duration: Nov'20 – Aug'22
Nov'20 to Aug'22 with LNT Infotech as Specialist Data Engineering

Key Result Areas: With Client - Travelers (Nov 2020 – Aug 2022)

- Development of data pipeline using SIP and orchestration using Autosys which picks data from different sources and lands into S3 raw regions for consumption.
- Development of staging and parsing framework which consist of reading data from S3 regions connectivity through Pyspark in Databricks environment and parsing them, loading into the Glue tables and query using Athena query engine.
- Involved in Prod support of existing SIP jobs and fixing all critical client P1 bugs.
- Involved in triggering Pyspark parsing jobs using Orchestration Terraform by invoking CI/CD Jenkins pipelines.
- Regular interaction with client for getting clear vision of requirement and refinement in requirement by demoing the developed segment of project.

Organization: WIPRO Ltd.
Duration: Jul'18 - Nov'20
Jul'18 to Nov'20 with WIPRO Ltd. as Technical Lead-Big Data

Key Result Areas: With Client - MITIE Mozaic (Feb'2019 - Nov'2020)

- Development of data pipeline using Streamsets so that data flow from IOT sensors to Kafka topics.
- Development of framework which consists of different jobs as per the business logic.
- Development of API using Python to expose data to the user dashboard.
- Involved in Fixing all critical client P1 bugs.
- Developed a data re-processing scripts/ code which fetches the data from Kafka in case of missed data and re-processed it further.
- Involved in the development of shell scripts for triggering spark jobs either to start or stop individual spark jobs in cluster.

Key Result Areas: with Client – Capital One (Jul'2018 - Feb'2019)

- Development of quantum sub workflow JSON for the Alexandria features.
- Development of common Batchconfigs YML for Groups involved which contains multiple features.
- Unit testing of the sub-workflow with snowflake data, by executing developed queries in Snowflake console, and verifying Unit results.

- Submitting Application on Spark Cluster in Client mode and validating output in S3 bucket.
- Computed data from AWS S3, and Snowflake using Spark Data frames and performed transformations and actions.
- Experienced in performance tuning of Spark Applications by doing serialization, correct level of Parallelism, caching, and memory / core tuning of executors and Broadcasts.
- Monitored the Spark History Server for the performance optimization of the spark DataFrames.

Organization: COGNIZANT TECHNOLOGY SOLUTIONS
Duration: Apr'16 - Jun'18
Apr'16 to Jun'18 with Cognizant Technology Solutions as Associate Projects-Big Data

Key Result Areas: With Client - Sabre (May' 2017 - Jun' 2018)

- Developed a Spark Streaming consumer application using Scala and Kafka Direct API for real-time data ingestion
- Built a custom Apache Flume interceptor to convert EBCDIC to ASCII for downstream processing compatibility
- Created a Maven-based Scala project to integrate reusable Java business logic into Spark workflows
- Designed and implemented shell scripts to start, stop, and monitor Spark Streaming applications in production
- Automated batch and streaming job orchestration using Oozie Coordinators, Bundles, and Workflows
- Configured Oozie Spark actions to schedule and execute Spark JAR jobs across the Hadoop cluster
- Monitored and tuned Spark applications using Spark History Server, improving job performance and resource utilization

Key Result Areas: With Client – ING Vyasa (Apr' 2016 - May' 2017)

- Identified datasets and defined metadata based on business requirements
- Designed end-to-end data ingestion and standardization frameworks in Hadoop ecosystem
- Planned HDFS file formats, compression strategies, directory structures, and partitioning approaches
- Designed Hive table schemas and partition structures for raw, standardized, and reference datasets
- Developed batch file–based and Sqoop-based ingestion jobs for initial and incremental data loads
- Managed data landing into HDFS staging/landing zones using Hadoop FS commands
- Built Oozie workflows to automate ingestion and downstream data processing based on required schedules
- Landed and maintained reference data in HDFS common zones for enterprise consumption
- Designed and implemented data standardization processes, including schema enhancements and additional reference columns
- Created HiveQL scripts to:
  - Standardize and enrich transaction data using reference datasets
  - Move validated data from Landing Zone to Standardized Zone
  - Process incremental transaction updates and apply reference data augmentation
  - Automated end-to-end data pipelines by integrating ingestion, reference loading, and standardization workflows through Oozie.

Organization: DELL INTERNATIONAL SERVICES
Duration: Jan'13 - Apr'16
Jan'13 to Apr'16 with Dell International Services as Software Dev Analayst.

Key Result Areas: With Client - Sabre (Mar' 2015 - Apr' 2016)

- Import the Transactional Data directly from SQL Server into HDFS using Sqoop.
- Perform text analytics to annotate the Transactional Data using Apache UIMA and JAVA.
- To store the annotated Data in HBase in JSON Format.
- The stored data is used by various components of the solution such as Tx Pattern Data Mining, Rule Engine, Predictive Scoring and Offer Rotation Engine.
- Offer Rotation Engine selects and delivers offers through SMS or Email.

ACADEMIC DETAILS & ACHIEVEMENTS

- B.E. Computer Science with First Class in 2012 from Nagpur University,India
- Two times awarded with Dell Services – On the Spot Award for outstanding performance and lasting contribution
- Have been awarded the "Quarter of the year" by Cognizant.